# TOMMASO CRIPPA

## High Performance Computing Engineering Student

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*HPC Engineering student with hands-on experience building data pipelines and ML-backed analytics tools. Proficient in Python, SQL, and Kafka, seeking to apply these skills in scalable data infrastructure and ETL processes.*

## EDUCATION

**Double Master's Degree in High Performance Computing** — **Politecnico di Milano - University of Luxembourg**
29.25 / 30 — 2024 - present

- Selected for the exclusive EUMaster4HPC double-degree programme, in which I participated in extracurricular HPC challenges, an international summer school and workshops with industry leaders

**Bachelor degree in Computer Engineering** — **Politecnico di Milano**
110 cum laude / 110 — 2021 - 2024

- Won "Migliori Matricole" scholarship for being one of Politecnico di Milano's best freshmen students of the 21/22 season.

*Relevant courses completed:* Streaming Data Analytics, Time Series Analysis, Artificial Neural Networks and Deep Learning

## EXPERIENCES

**Master's Thesis Student @ MEGWARE Computer Vertrieb und Service GmbH** — **Chemnitz**, Germany
HPC | Machine Learning | Time-Series | SLURM | Linux — February 2026 - present

Crippius/hpc-bottleneck-detector

- Building a deployed analytics service that ingests ∼5,000 data points per minute from 100+ HPC metrics, applies ML-based anomaly detection, and delivers actionable bottleneck reports to HPC users.
- Designing the end-to-end data architecture: from the initial data ingestion and transformation to the automated ml pipeline to enable optimization recommendations without source-code instrumentation.

**Operations & Logistic @ Google Developer Group On Campus - Polimi** — **Milan**, Italy
AI | Cloud | Mobile — August 2024 - August 2025

- Organised and executed 5+ technical events over 12 months on AI, Cloud Computing, and Mobile Development, including a hackathon with 150+ participants, doubling community membership over the year.

## PROJECTS

**Benchmarking Detectors in Streaming Machine Learning**
Python | capymoa | pandas | numpy | matplotlib | Colab — Streaming-Data-Analytics/Benchmarking-Detectors-in-SML

- Benchmarked concept drift detectors across 4 streaming datasets (∼3M data points total) covering abrupt, gradual, incremental, and recurrent drifts, evaluating accuracy, Kappa, and runtime to identify best-fit detectors per scenario.

**Active Learning for Social Media Data Classification**
Python | tensorflow | matplotlib | numpy | pandas | Colab — Crippius/al_project

- Built an active learning pipeline selecting the most informative tweets from 325,000 posts using different selection techniques, reaching F1 ∼0.8 with only 1,000 labels available through hyperparameter tuning.

**Whatsapp Wrapped**
Python | matplotlib | pandas | fpdf | Flask | HTML | CSS | Javascript | Vercel | Render — Crippius/whatsapp-wrapped

- Built a full-stack web app that transforms WhatsApp chat exports into personalised analytics reports (message trends, word frequency, response times), deployed on Vercel and Render with 100+ users since launch.

**Early Alzheimer Detection with Deep Learning**
Python | pytorch | keras | nilearn | C++ | libtorch | clinica | SLURM — justvicthor/Early_Alzheimer_Detection_DL

- Developed a deep learning model for early Alzheimer's detection, achieving Macro-AUC of 0.9197 on the test set, improving upon the original paper (0.80), with full data preprocessing and augmentation pipelines run on the MeluXina supercomputer.

## SKILLS AND LANGUAGES

**Tech Stack:** Python (Advanced), Java, Kafka, SQL (Advanced), Spark
**Infrastructure:** Git (Advanced), DVC, Docker, AWS
**Languages:** Italian (Native), English (Proficient – IELTS 7.5), German (Beginner – A2)